

Sydney Bank Customer Churn



Group E

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Executive Summary

Help Sydney Bank focus on customers most likely to leave



Who to contact

160

customers to contact

Focus on inactive, newer and low-activity customers



What we found

~3x

better than random selection

19 of 32 selected customers actually left in our test



What to do

3 months

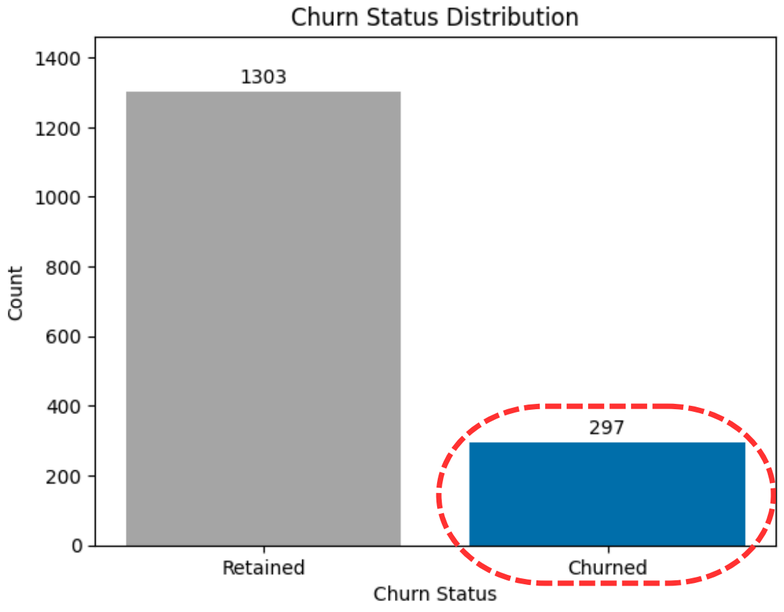
to test approach

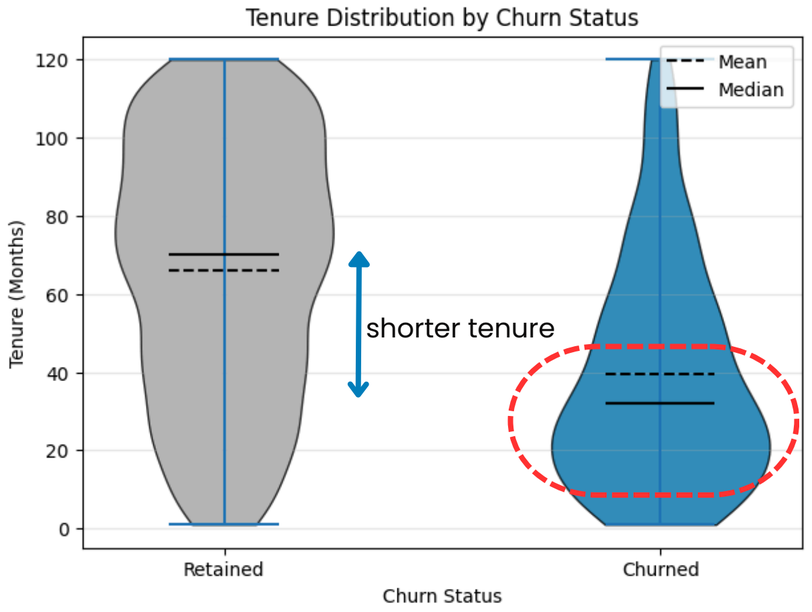
Try re-engagement offers, early check-ins and activity reminders

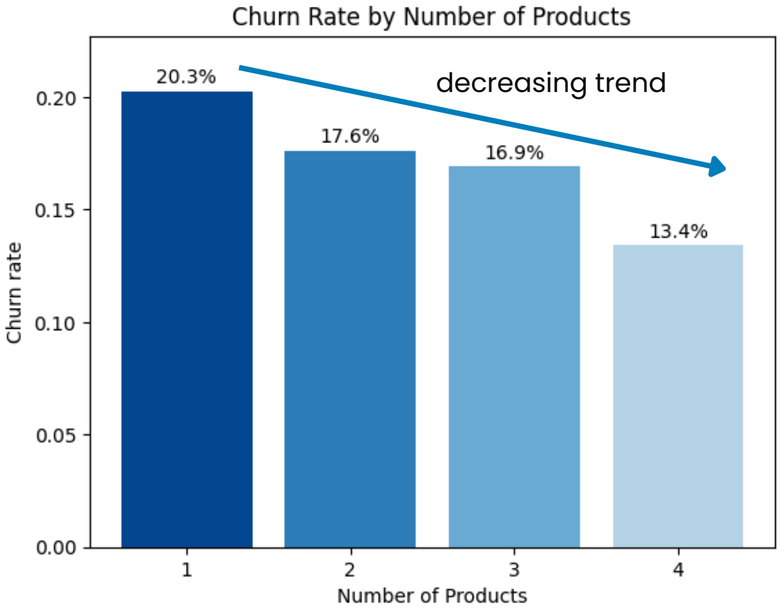


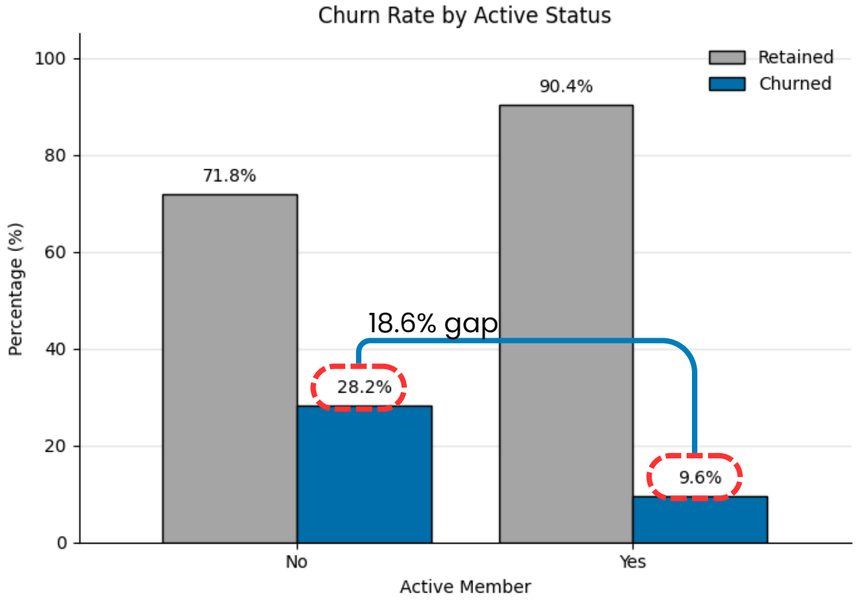
Compare with usual service to see how many extra customers stay

Customer Landscape









18.6%

Customer churned

Hypotheses

1. Short-term customers with low transaction activity may have a higher risk of churn

2. Churned customers tend to have shorter tenure.

3. Short-term customers with inactivity and low transaction activity may represent a particularly high churn risk

Churn Prediction Plan

Target

1 = Churn 0 = Retain

Output: Churn Probability

8

Candidate Features

excluding CustomerID

80%

Train set

20%

Test set

Data Preprocessing

Data cleaning

standardized inconsistent Gender values

Pipeline

Categorical Features

encoded using One-hot encoding

Numerical Features

scaled using StandardScaler

Model fitting

Churn probability

Decision Tree does not need scaling

Models and Evaluation Metrics

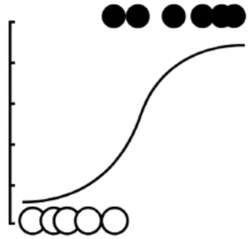
Models

baseline model



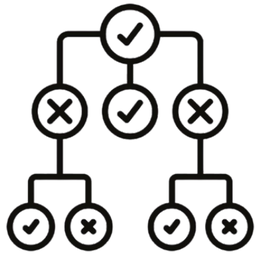
Majority Class

predict everyone as retained



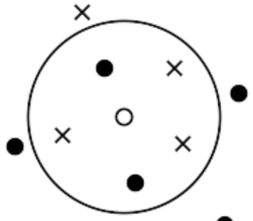
Logistic Regression

simple, interpretable risk scores



Decision Tree

capture non-linear patterns



k-Nearest Neighbor

similar-customer behavior

Evaluation

Stratified 5-fold cross-validation

- Preserve churn proportions across folds.
- Tune full pipelines, then evaluate on the 20% test set.

Precision at top 10% (primary)

- Share of actual churners among priority customers.
- Select models that make the limited contact capacity most effective.

ROC-AUC (support)

- How well the model ranks churners above retained customers.

Why not accuracy?

- High accuracy can be achieved by labelling customers all as retained but captures no churners.

Baseline & Logistic Regression

Baseline

predict all customers as retained

81.4%

Accuracy

0%

Churn captured

No useful customer risk ranking

Logistic Regression

C = 1

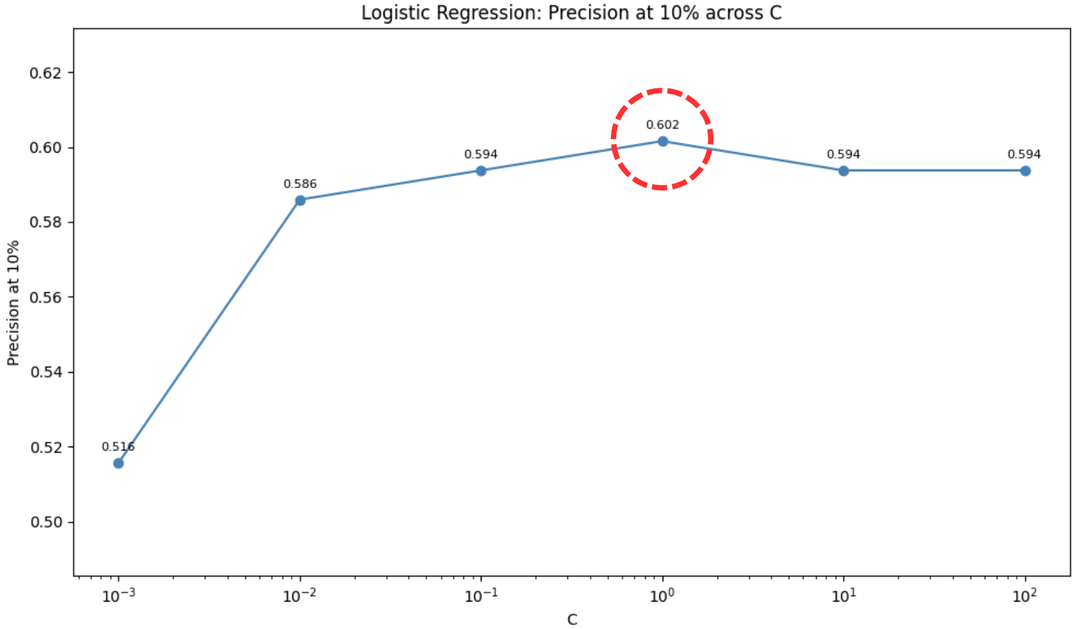
selected settings

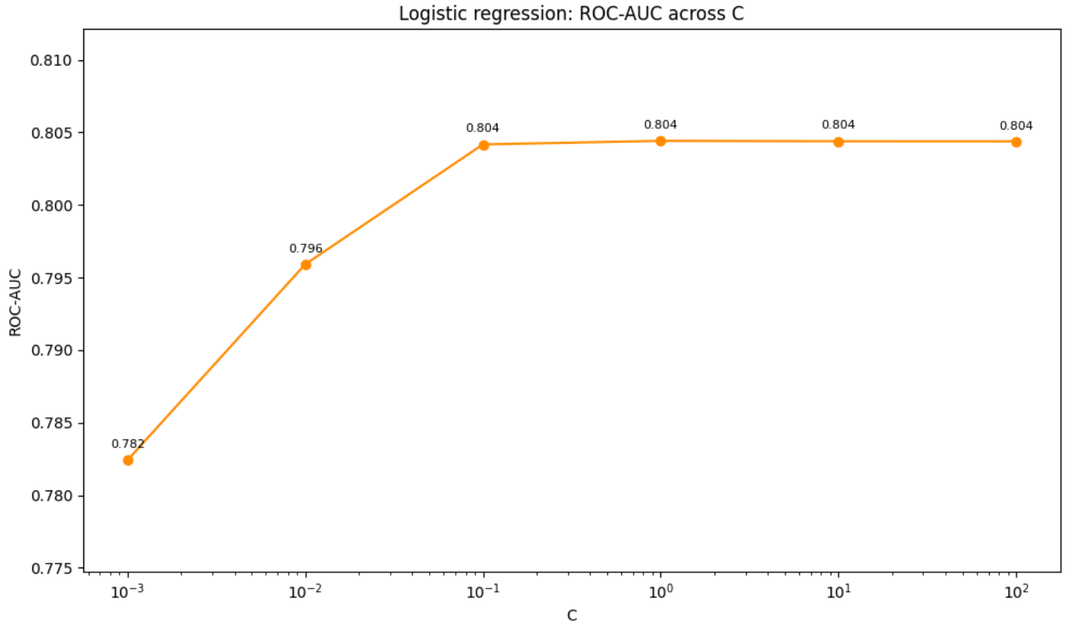
60.2%

CV Precision at 10%

0.804

ROC-AUC





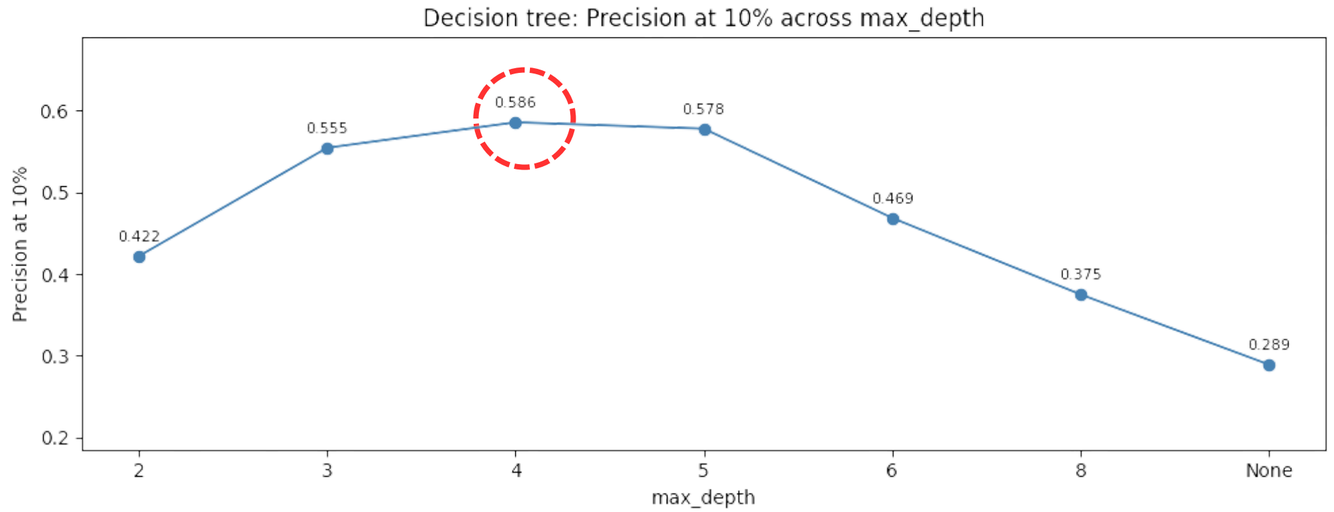
Decision Tree & KNN

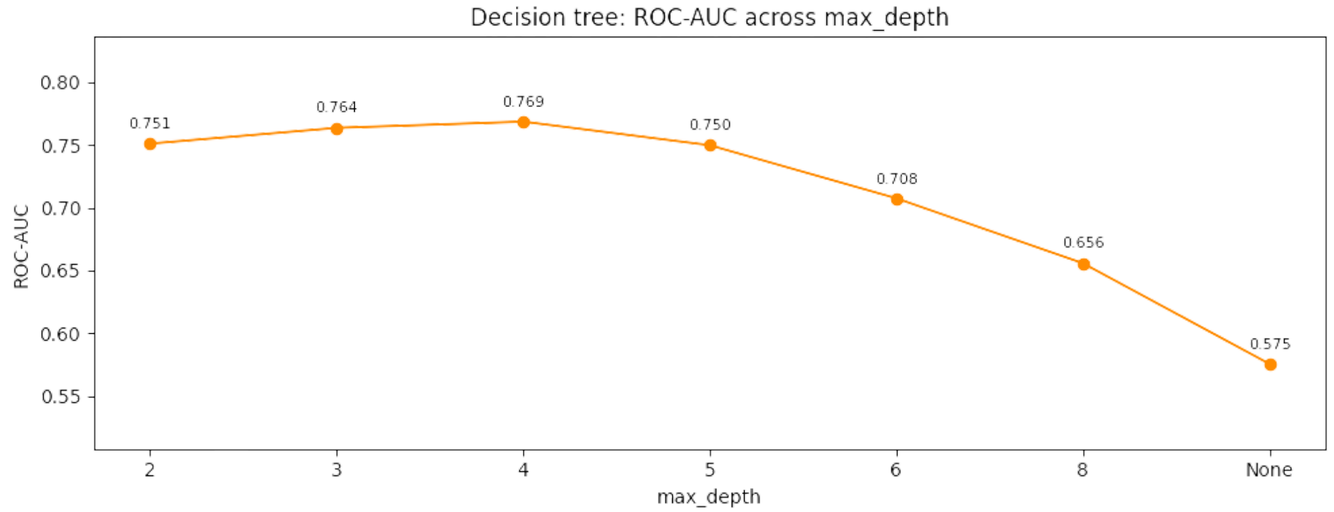
Decision Tree

4
maximum depth

58.6%
CV Precision at 10%

0.769
ROC-AUC



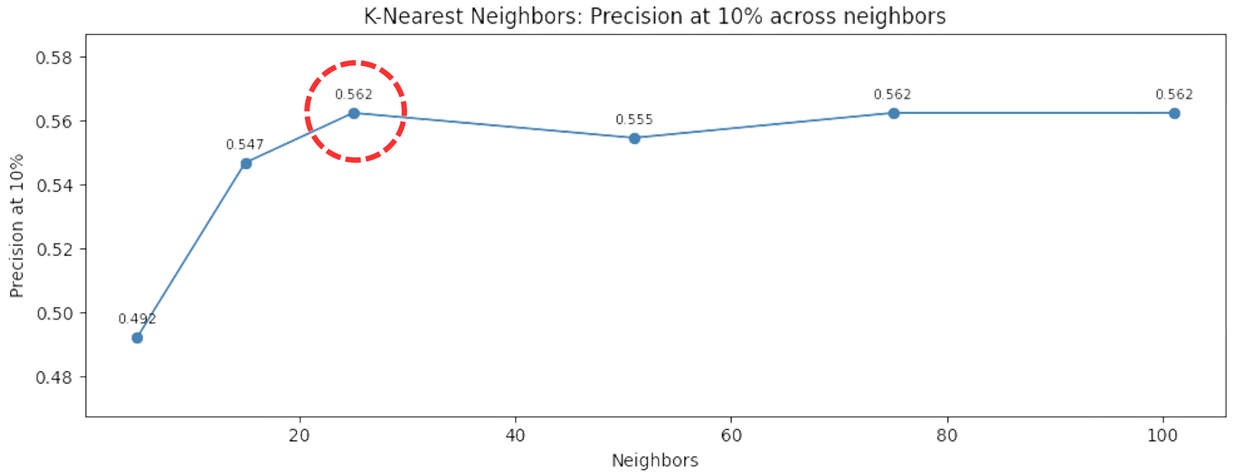


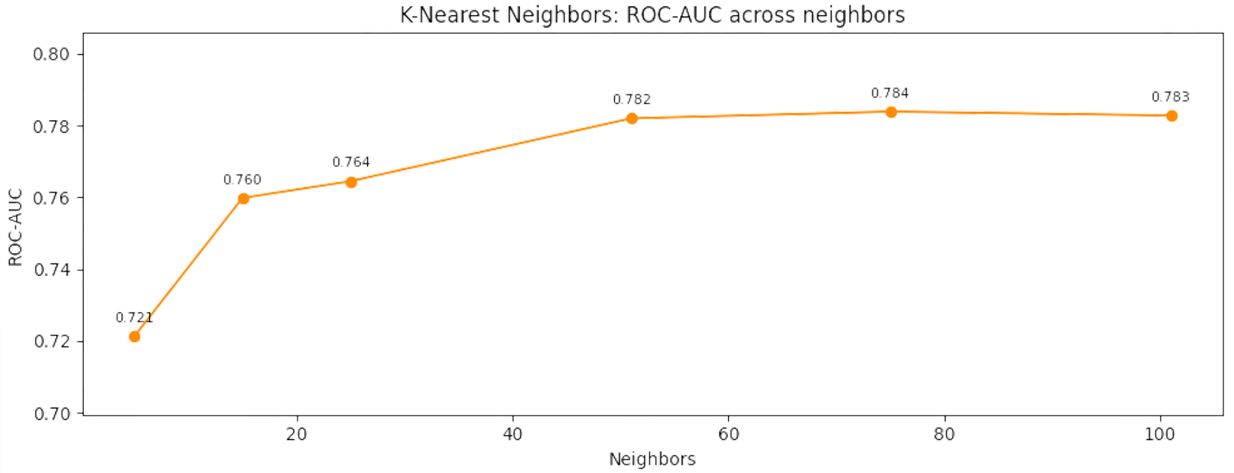
K-Nearest Neighbors

25
neighbors

56.2%
CV Precision at 10%

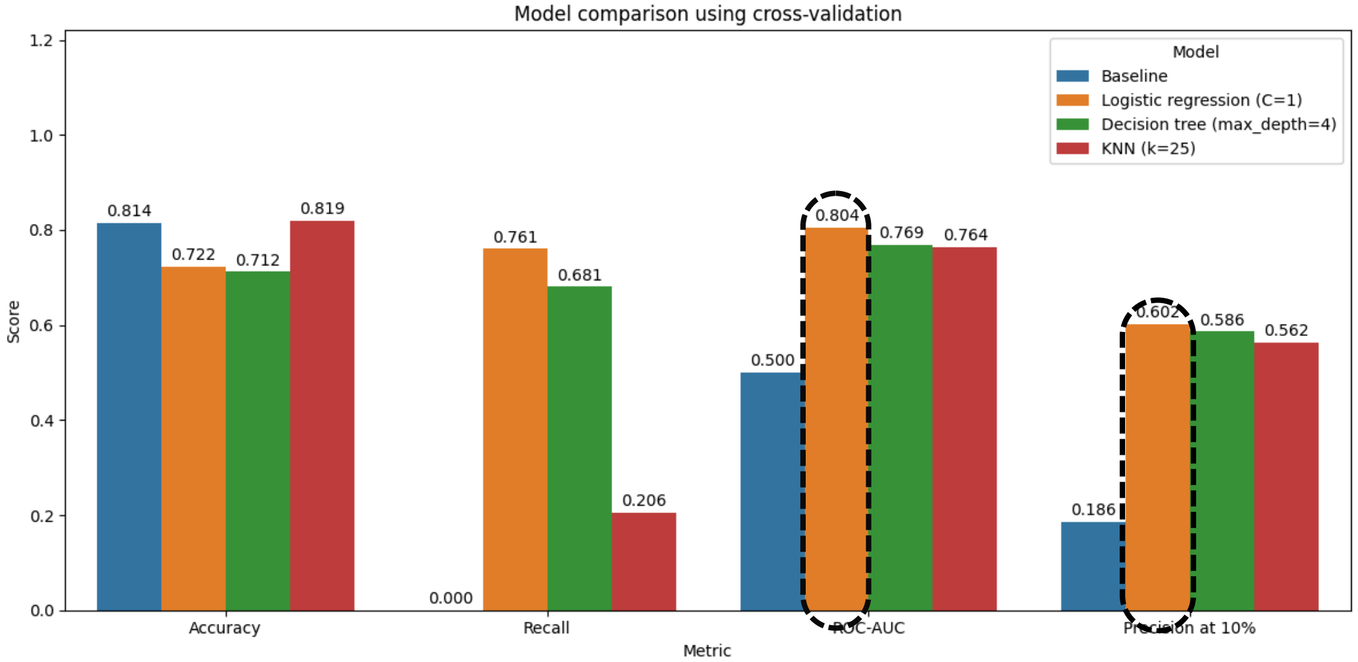
0.764
ROC-AUC





Model Selection & Evaluation

Model Selection



Logistic Regression standouts in metrics we chose

Model Evaluation

59.4%
Precision at top 10%

0.782
ROC-AUC

71%
Accuracy

Among 32 customers we contact, 19 are churners

Final Model

$$Pr(\text{Churn}) = \frac{1}{1 + e^{-z}}$$

$$z = 0.286 - 1.007T - 0.311B - 0.313N - 1.474A$$

where:

- T: standardised tenure
- B: standardised balance
- N: standardised transactions in the last six months
- A: active membership (1 = Yes, 0 = No)

Recommendation & Limitation

Recommendations

Inactive customers

Offer a target re-engagement incentive

Short tenure

Flag customers with lower than 40 months

Low Transaction Activity

Flag low transaction activity in 6 months

Limitations

Risk indicators do not prove the causes

Benefits of retained customer remain untested

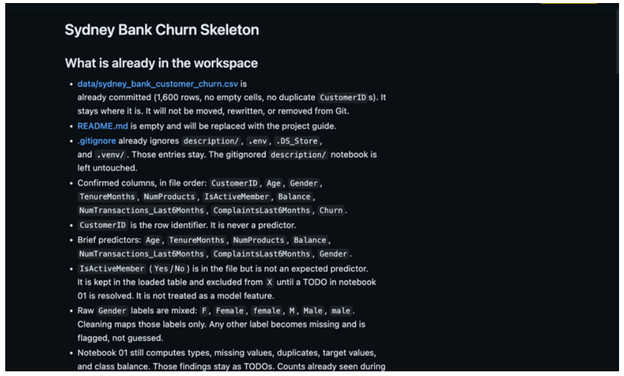
Next step: controlled pilot

Test whether recommended actions improve retention before scaling up



Generative AI Use

How we used

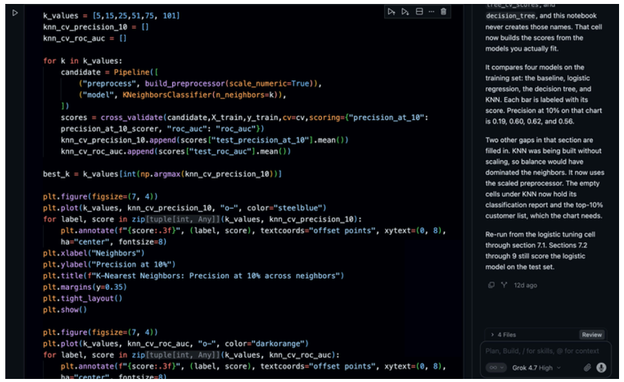


Sydney Bank Churn Skeleton

What is already in the workspace

- data/sydney_bank_customer_churn.csv is already committed (1,600 rows, no empty cells, no duplicate `CustomerID`s). It stays where it is. It will not be moved, rewritten, or removed from Git.
- README.md is empty and will be replaced with the project guide.
- .gitignore already ignores `description/`, `.env`, `.DS_Store`, and `.venv/`. Those entries stay. The gitignored `description/` notebook is left untouched.
- Confirmed columns, in file order: `CustomerID`, `Age`, `Gender`, `TenureMonths`, `NumProducts`, `IsActiveMember`, `Balance`, `NumTransactions_Last6Months`, `ComplaintsLast6Months`, `Churn`.
- `CustomerID` is the row identifier. It is never a predictor.
- Brief predictors: `Age`, `TenureMonths`, `NumProducts`, `Balance`, `NumTransactions_Last6Months`, `ComplaintsLast6Months`, `Gender`.
- `IsActiveMember` (`Yes`/`No`) is in the file but is not an expected predictor. It is kept in the loaded table and excluded from `X` until a TODO in notebook 01 is resolved. It is not treated as a model feature.
- Raw `Gender` labels are mixed: `F`, `Female`, `female`, `M`, `Male`, `male`. Cleaning maps those labels only. Any other label becomes missing and is flagged, not guessed.
- Notebook 01 still computes types, missing values, duplicates, target values, and class balance. Those findings stay as TODOs. Counts already seen during

Plan skeletons of the project



```
k_values = [5,15,25,51,75, 101]
knn_cv_precision_10 = []
knn_cv_roc_auc = []

for k in k_values:
    candidate = Pipeline([
        ("preprocess", build_preprocessor(scale_numeric=True)),
        ("model", KNeighborsClassifier(n_neighbors=k)),
    ])
    scores = cross_validate(candidate,X_train,y_train,cv=cv,scoring={"precision_at_10":
    precision_at_10_scorer, "roc_auc": "roc_auc"})
    knn_cv_precision_10.append(scores["test_precision_at_10"].mean())
    knn_cv_roc_auc.append(scores["test_roc_auc"].mean())

best_k = k_values[int(np.argmax(knn_cv_precision_10))]

plt.figure(figsize=(7, 4))
plt.plot(k_values, knn_cv_precision_10, "o-", color="steelblue")
for label, score in zip(tuple[int, Any](k_values, knn_cv_precision_10):
    plt.annotate(f"{score:.3f}", (label, score), textcoords="offset points", xytext=(0, 8),
    ha="center", fontsize=8)
plt.xlabel("Neighbors")
plt.ylabel("Precision at 10%")
plt.title(f"K-Nearest Neighbors: Precision at 10% across neighbors")
plt.margins(y=0.35)
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 4))
plt.plot(k_values, knn_cv_roc_auc, "o-", color="darkorange")
for label, score in zip(tuple[int, Any](k_values, knn_cv_roc_auc):
    plt.annotate(f"{score:.3f}", (label, score), textcoords="offset points", xytext=(0, 8),
```

decision_tree, and this notebook never creates those names. That cell now builds the scores from the models you actually fit.

It compares four models on the training set: the baseline, logistic regression, the decision tree, and KNN. Each bar is labeled with its score. Precision at 10% on that chart is 0.19, 0.60, 0.62, and 0.56.

Two other gaps in that section are filled in. KNN was being built without scaling, so balance would have dominated the neighbors. It now uses the scaled preprocessor. The empty cells under KNN now hold its classification report and the top-10% customer list, which the chart needs.

Re-run from the logistic tuning cell through section 7.1. Sections 7.2 through 9 still score the logistic model on the test set.

12d ago

4 Files Review

Plan, Build, / for skills, @ for context

Grok 4.7 High

Build the project

How we checked



Reviewed the code against the actual dataset

Reviewed the formulas and interpretations

Checked actions against assignment requirements

Our responsibility



Made analytical and technical decisions

Remained responsible for the accuracy of the model

Acknowledged how GenAI supported the project

Generative AI helps us plan and build the project. Our responsibilities are checking the accuracy of the build and modify the plan

Thanks for listening!

Any questions?